RiskLens

News-Driven Risk Intelligence & Event-Triggered Portfolio Stress Testing

Turning fragmented public information into an inspectable risk-analysis workflow.



Illustrative mockup · layout only

S&P Global & CRISIL Campus Hackathon 2026 · Kinjal Gupta · Computational Data Science

THE PROBLEM · WHY THIS MATTERS

Risk signals are scattered. Portfolio consequences are harder to inspect.

News feeds
Fragmented headlines

GAP

Risk interpretation
Which exposures matter?

GAP

Portfolio scenario
Hard to inspect

1

Relevant risk signals appear across news sources and market data.

2

A headline alone does not explain which portfolio exposures may be vulnerable.

3

Analysts need a traceable workflow to inspect events and explore possible scenarios.

From headline overload to structured risk exploration.

One workflow from incoming news to scenario analysis

1

Ingest

Collect public news from GDELT and RSS.

2

Interpret

Clean and deduplicate records; calculate sentiment and assign keyword-based event categories.

3

Inspect

Explore headlines, sources, timestamps and risk indicators in a Streamlit dashboard.

4

Stress-test

Apply user-selected hypothetical shocks to a synthetic portfolio.

SEPARATE MARKET REFERENCE View historical NIFTY 50 prices and daily returns fetched through Yahoo Finance.

VADER sentiment and keyword event labels are heuristics, not a validated AI prediction model.

ARCHITECTURE & TECHNOLOGY

Modular by design. Transparent about assumptions

NEWS INTELLIGENCE PIPELINE

DATA SOURCES

GDELT DOC API · Public RSS feeds

→

PROCESSING

Pandas cleaning and deduplication · VADER sentiment · Keyword-based event classification · Heuristic impact scoring

→

DASHBOARD

Streamlit · Plotly · News table and risk visualizations

MARKET-DATA REFERENCE

MARKET DATA

Yahoo Finance via yfinance

→

NIFTY 50 REFERENCE

NIFTY 50 historical closing prices and daily returns

IMPORTANT DISTINCTION

The NIFTY 50 market-data component is separate from the hypothetical portfolio-shock engine. Market returns do not calibrate the scenario shocks in the current implementation.

PORTFOLIO STRESS TEST

INPUTS

Synthetic ₹100 Cr portfolio · Selected event scenario · Illustrative asset shocks

→

OUTPUTS

Simulated portfolio impact · Loss attribution by asset

Python | **Streamlit** | **Pandas** | **Plotly** | **VADER Sentiment** | **feedparser** | **Requests** | **yfinance**

Event labels, impact scores, and asset shocks are not validated predictions.

DATA & DEMONSTRATION

Real public data · Synthetic portfolio assumptions

REAL DATA SOURCES

- Public news metadata retrieved from GDELT and RSS feeds.
- NIFTY 50 historical closing prices and daily returns retrieved through Yahoo Finance.
- Available source URLs and ingestion provenance retained.

ILLUSTRATIVE INPUTS

- Synthetic ₹100 crore portfolio across five asset categories.
- User-selectable scenario shocks and heuristic impact scores.
- Simulated losses used to explore hypothetical portfolio sensitivity.

SYNTHETIC PORTFOLIO EXPOSURES	Exposure
Corporate loans	₹30 Cr
Government bonds	₹20 Cr
Banking equities	₹15 Cr
Infrastructure loans	₹20 Cr
Synthetic derivatives exposure	₹15 Cr
Total	**₹100 Cr**

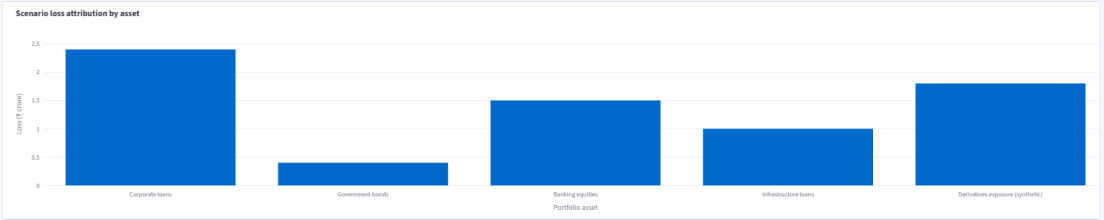
NIFTY 50: Historical prices and daily returns.



News analysis: Sentiment and heuristic impact.



Portfolio stress test: Simulated loss attribution by asset.



REPRODUCIBILITY: A NIFTY 50 historical CSV snapshot and a synthetic portfolio CSV are included in `/data`.

Sentiment and impact scores are heuristic. Portfolio losses are simulated under illustrative assumptions and are not forecasts or measured losses.

A foundation for faster, more transparent risk exploration

POTENTIAL VALUE

- Brings multiple public information sources into one workflow.
- Makes the path from a headline to scenario assumptions easier to inspect.
- Allows users to explore how hypothetical shocks affect different portfolio exposures.

CURRENT LIMITATIONS

- Event categories are keyword-based; sentiment is calculated using VADER.
- Impact scores and asset shocks are heuristic assumptions.
- No validated causal relationship between news and market movements has been established.
- Results are not forecasts or measured portfolio losses.

NEXT STEPS

- Validate event classifications against manually labelled data.
- Calibrate scenario shocks using documented historical episodes.
- Add automated tests and systematic source-failure handling.

RiskLens — See the signal. Inspect the exposure. Test the scenario.